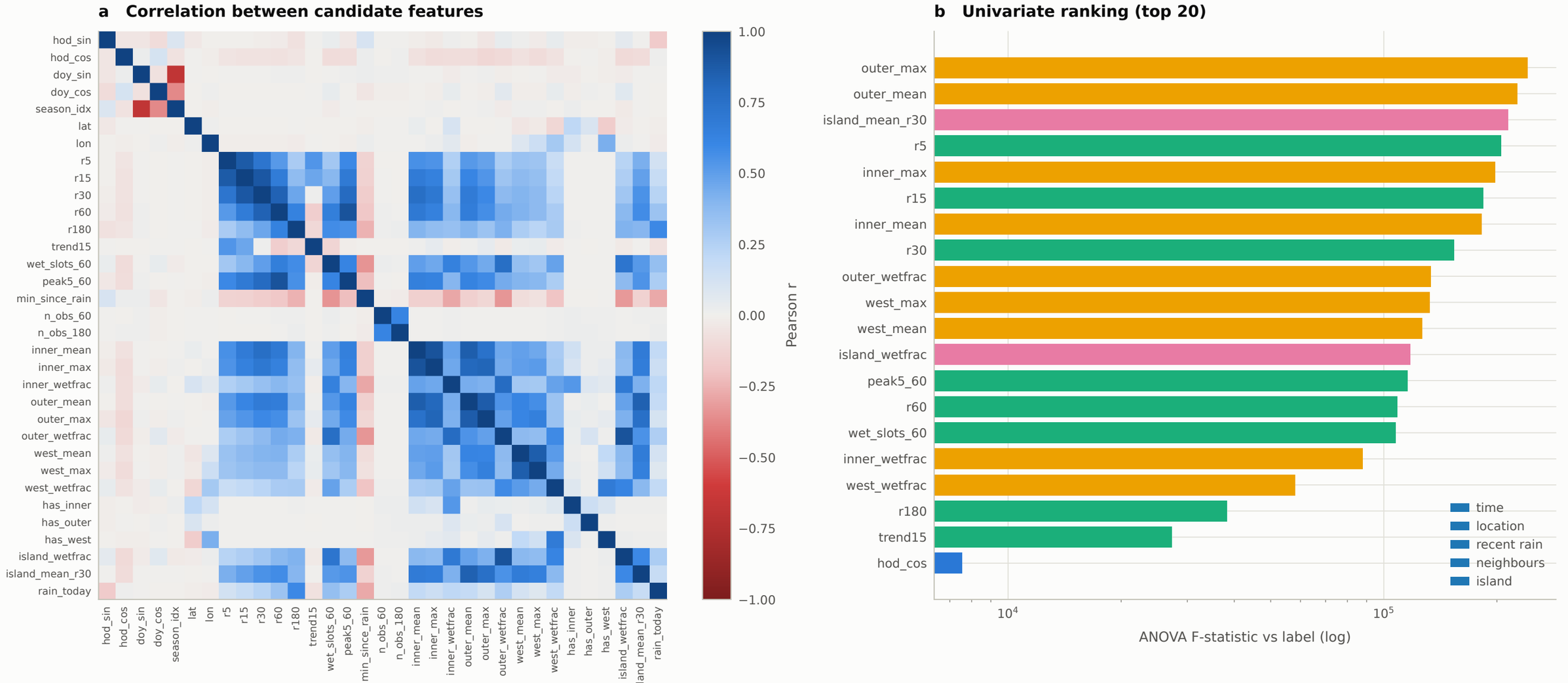
Recent rain at the station and its neighbours carries most of the signal

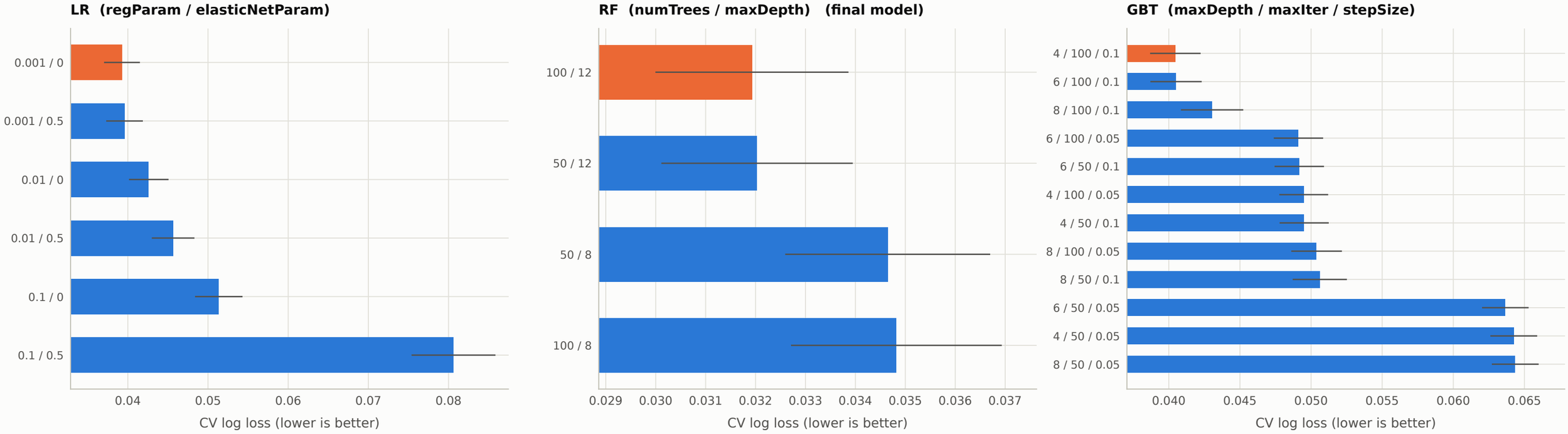
Left: several recent-rain windows are strongly correlated (they overlap in time). Right: ranking by the ANOVA F-test (label 0 vs 1), coloured by feature group.



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.

The random forest has the lowest cross-validated log loss and becomes the final model

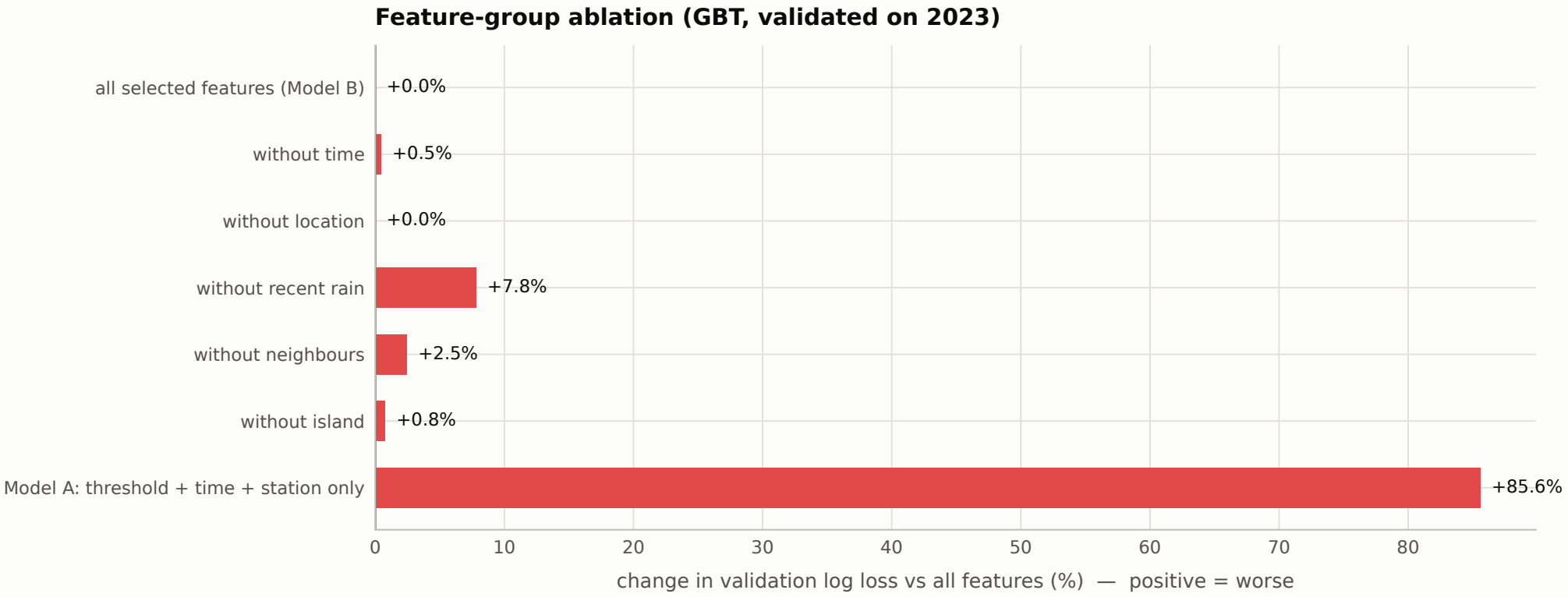
Each bar = mean weighted log loss over 4 year-blocked folds (error bar = s.d. across folds). Orange = best setting per model.
Best settings lie on the edge of the grid (RF: numTrees, maxDepth; GBT: maxDepth, maxIter, stepSize), so a wider grid might improve them further.



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.

Removing the recent rain features hurts most; time and station alone are far weaker

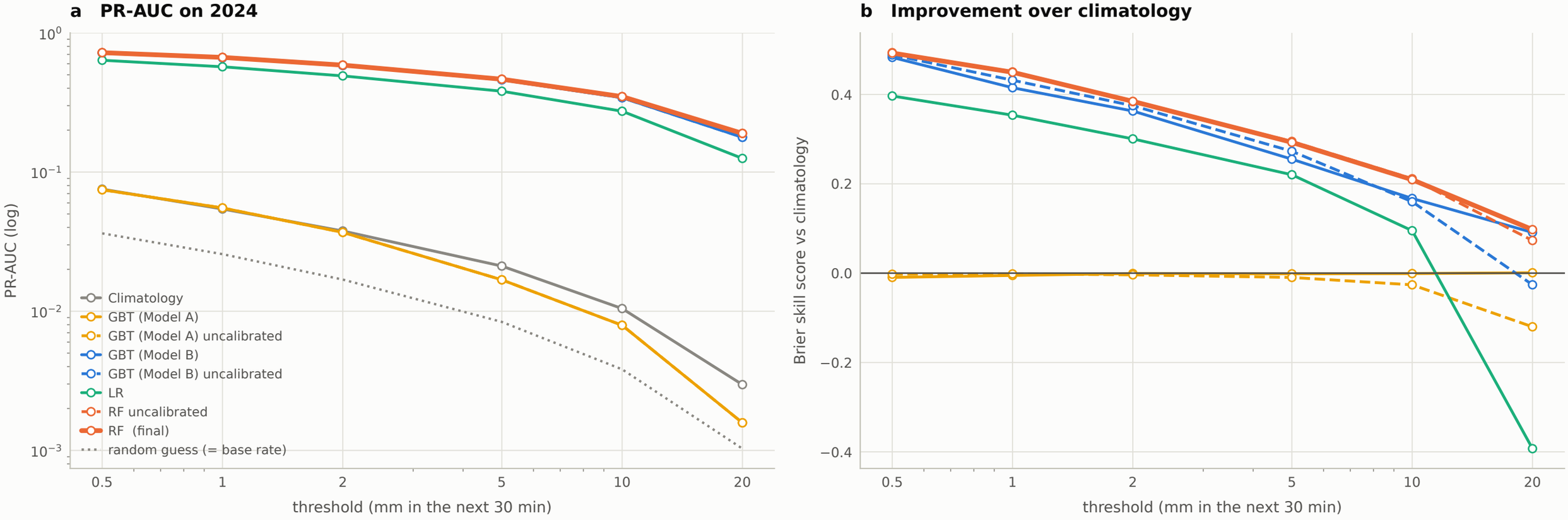
Each bar retrains the model without one feature group. Model A uses only what is known without any rain observation.



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.

On unseen 2024 data the random forest removes 29% of the climatology error at 5 mm

PR-AUC at 5 mm = 0.46 against a base rate of 0.0084 (55× better than random). Model A (no rain observations) is no better than climatology.

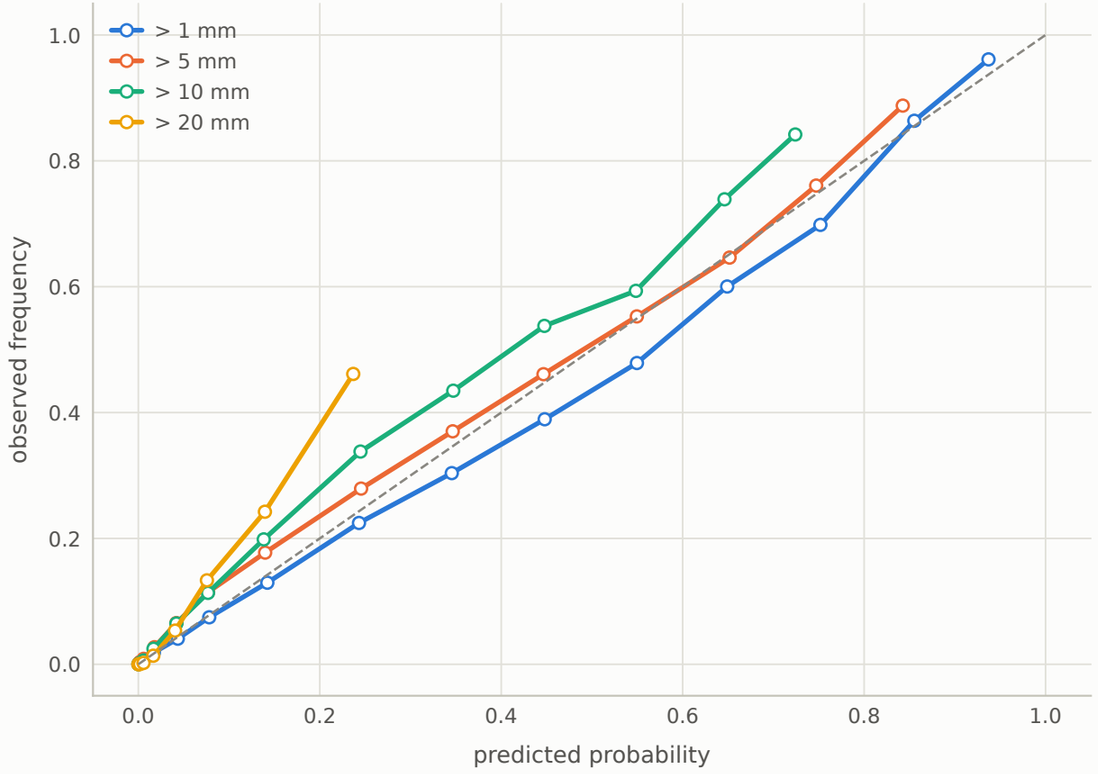


Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.

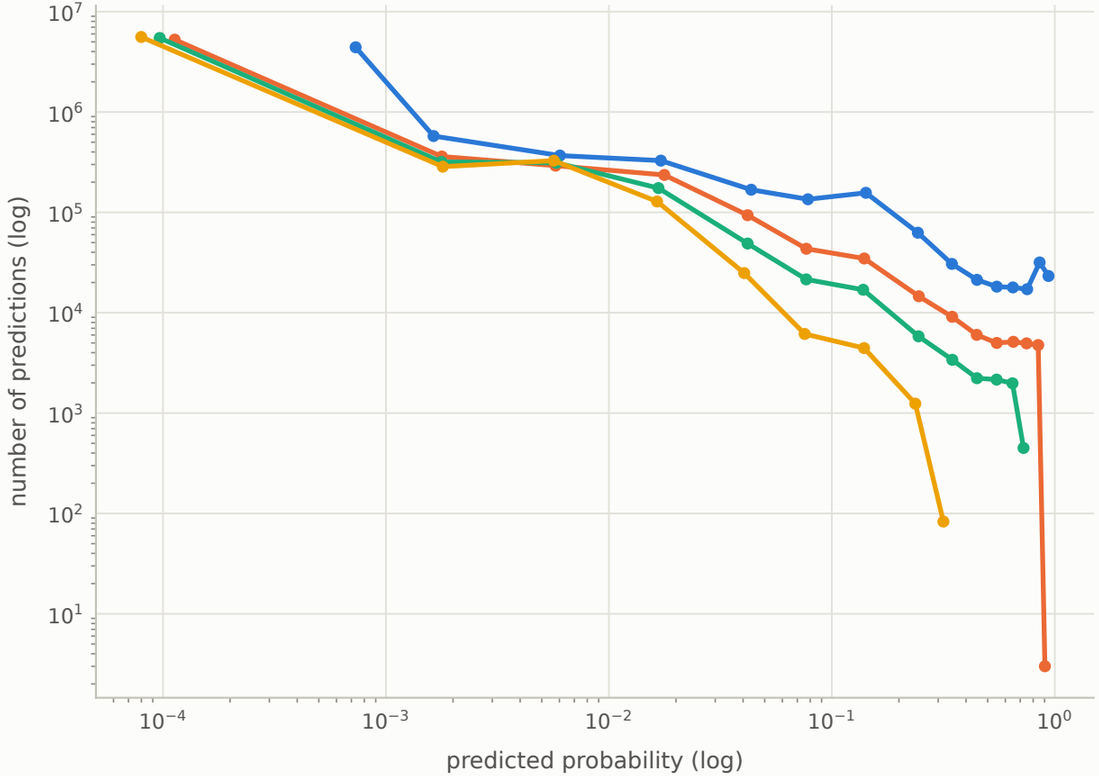
Probabilities below 10% are reliable; the reliability of high forecasts depends on the threshold

Below 10%, where almost all forecasts lie, observed frequency differs from the forecast by +0.001 on average. Above 20%, events happen 3 points less often than forecast at 1 mm and 9 points more often at 10 mm. Expected calibration error at 5 mm = 0.0017.

a Reliability diagram (RF, 2024)



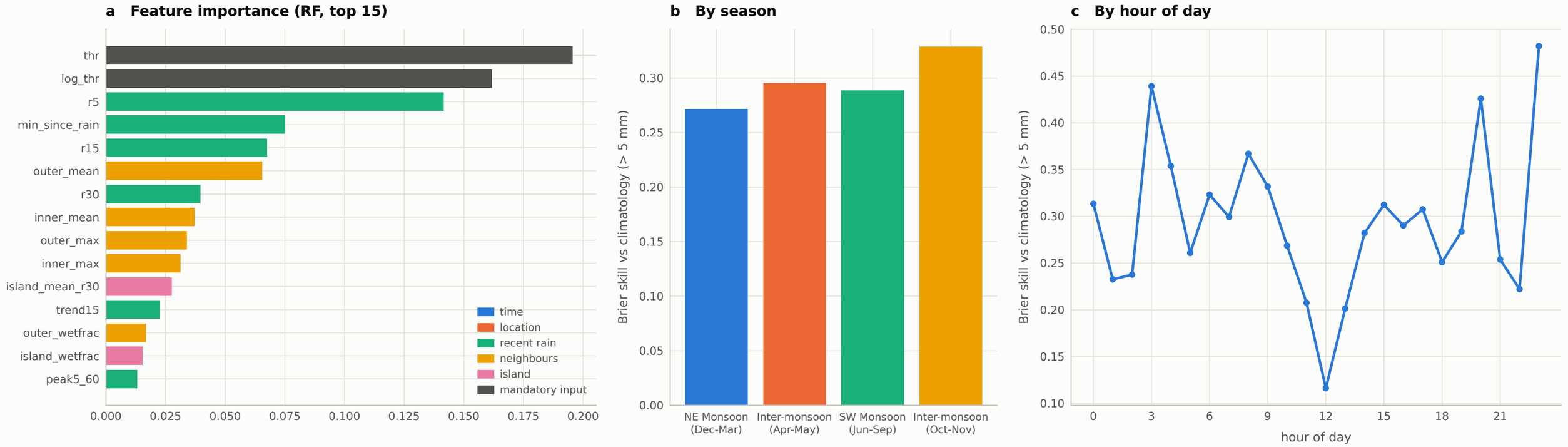
b How often each probability is predicted



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.
Bins with fewer than 100 predictions are not shown in panel a.

Recent rain nearby drives the forecast; skill is highest in the Inter-monsoon (Oct-Nov) and lowest around 12:00

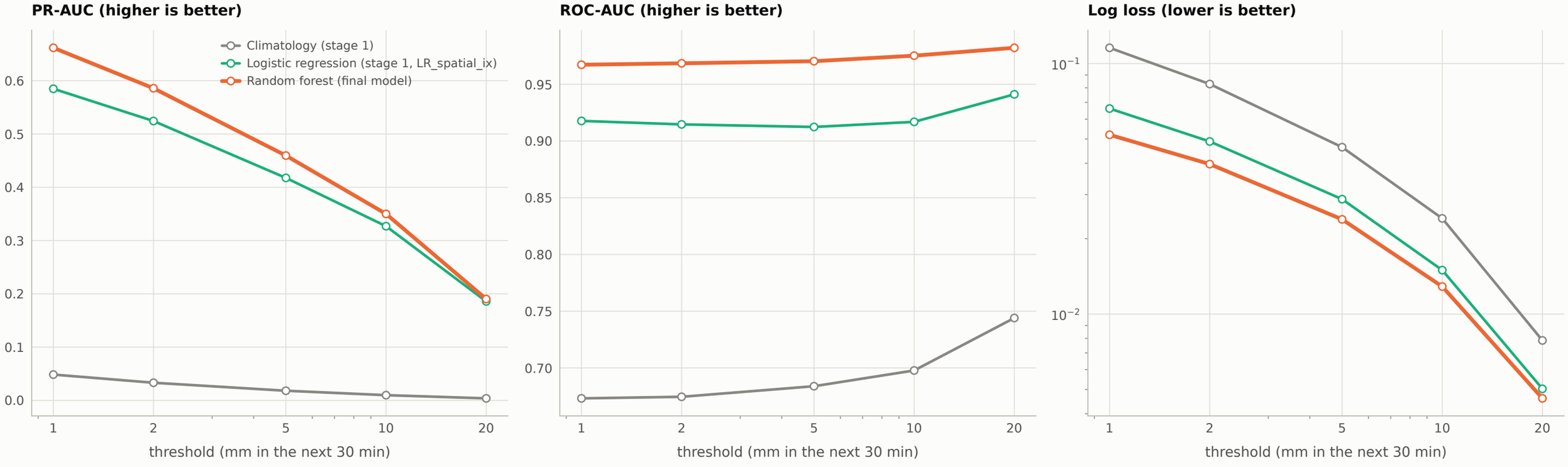
Importance = mean decrease in impurity (Spark featureImportances). Skill = 1 − Brier(model) / Brier(climatology) on 2024 for the 5 mm threshold.



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.

On the same 2024 rows the random forest beats the Stage 1 logistic model (PR-AUC at 5 mm 0.46 vs 0.42)

6,275,015 matched station-time-threshold rows (the Stage 1 evaluation rows). Same labels, same rows; only the models differ.



Source: data.gov.sg rainfall 2017–2024; Spark MLlib. Final model: selected by 4-fold year-blocked cross-validation on 2017–2023, tested once on 2024.